



Tennessee TECH

Default Question Block

CSC 1300 & CSC 1310 **Program Report** **Fall 2025**

You will use this **same link** for every programming assignment in CSC 1300 & CSC 1310 this semester.

TTU Email Address:

What class & section are you in?

CSC 1300-001 (MWF 9-9:50am) GOLD group
CSC 1300-001 (MWF 9-9:50am) PURPLE group
CSC 1300-002 (MWF 11-11:50am) GOLD group
CSC 1300-002 (MWF 11-11:50am) PURPLE group
CSC 1310-001 (TR 1:30-2:45pm)
CSC 1310-002 (TR 9:30-10:45am)

Which programming assignment are you reporting on?

Program 1
Program 2
Program 3
Program 4
Program 5

How many estimated total hours did you spend on this programming assignment?

- 0-2 hours
 - 2-5 hours
 - 6-10 hours
 - 11-15 hours
 - 16-20 hours
 - 21-30 hours
 - more than 30 hours
-

**How early did you START on the assignment?
(Please be honest - this will not affect your grade in any way)**

- the week it was assigned
 - two weeks prior to due date
 - one week prior to due date
 - 3 days prior to due date
 - 2 days prior to due date
 - 1 day prior to due date
 - on the due date
 - the day after the due date
 - two days after the due date
-

What is something that you learned from this assignment?**What was the most challenging aspect of the assignment?****Indicate the extent to which you agree or disagree for the following statements.**

	Strongly Disagree	Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Agree	Strongly Agree
							
I enjoyed this assignment.	☐	☐	☐	☐	☐	☐	☐
I learned from this assignment.	☐	☐	☐	☐	☐	☐	☐
I think I made a good grade on this assignment.	☐	☐	☐	☐	☐	☐	☐

CSC 1310 Questions on Parallelism

Data parallelism _____.

divides the program's logic into independent tasks that can be executed on different processors simultaneously (in parallel).

divides data into chunks that are processed simultaneously (in parallel) using the same operation.

is a form of parallel execution where different tasks are performed concurrently on multiple processors.

uses multiple processors to each run the entire dataset independently to compare results.

Task parallelism _____.

is a form of parallel execution where different tasks are performed concurrently on multiple processors.

is the process of distributing work evenly across multiple processors to ensure that all computing resources are utilized efficiently and no processor is significantly more burdened than others.

divides data into chunks that are processed simultaneously (in parallel) using the same operation.

uses multiple processors to each run the entire dataset independently to compare results.

Which sorting algorithm is best suited for parallelization due to its divide-and-conquer nature?

Bubble sort

Selection sort

Merge sort

Insertion sort

Radix sort

Which type of parallelism involves applying the same task to different subsets of data?

Task Parallelism
Loop Unrolling
Data Parallelism
Thread Pooling
Pipelining
Load Balancing

What happens if you select more threads than the number of processor cores available on your system?

It increases efficiency and therefore runs faster.
It has no impact and only uses the number of threads available on your system.
It will likely slow down the execution time due to context switching.
It will cause a compilation error.

Which of the following best describes a thread in parallel programming?

A separate program running in its own memory space.
A variable that stores shared data.
A mini-program that shares memory and executes part of a task concurrently.
An error handler in C++ and other languages.

What is the primary goal of using OpenMP in C++ programs?

The goal is to reduce the number of lines of code.
The goal is to increase memory usage.
The goal is to speed up programs using multiple CPU cores (processors).
The goal is to prevent recursion overhead in algorithms.

What header file must you include to use the OpenMP library?

opmp.h
openm_parallel.h
openmp.h
omp.h

Which OpenMP directive is used to define a parallel region?

omp_set_num_threads()
#pragma omp parallel

```
#pragma omp critical  
#pragma omp
```

Which OpenMP function returns the unique ID of a thread?

```
omp_get_thread_num()  
omp_id()  
omp_get_thread_count()  
omp_thread_id()
```

True or False. Each thread in OpenMP has its own separate memory space.

Always true because each thread can only see the variables it is working with

Always false because variables are shared among threads and can't be declared private

False, but variables can be declared private so that each thread gets its own copy of the variable

None of the above

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